

9 (**long**) Binary data are received sequentially from an input I and the output Z becomes 1 if the sequence 010 is detected. The output Z is otherwise 0. The detector also responds to overlapping sequences, *i.e.*, the sequence 01010 should give $Z = 1$ after bit 3 is received and also after bit 5 is received.

(a) By considering the following four states

	Description
state 1	start
state 2	got 0
state 3	got 01
state 4	got 010

draw the state diagram for the system.

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(b) Two JK bistables, with outputs labelled as Q_A and Q_B respectively, are used to construct the detector. By using the bistable allocation below draw the state transition table.

	Q_A	Q_B
state 1	0	0
state 2	0	1
state 3	1	0
state 4	1	1

[10]

(c) Use Karnaugh maps to deduce simplified expressions for the connections between the bistables. Draw also the corresponding circuit diagram of the detector.

[12]

(d) State how many bistables would be needed if the detector is now designed to detect,

(i) the sequence 01 ,

[1]

(ii) the sequence 101 .

[1]